



HWA CHONG INSTITUTION
2022 C2 H2 CHEMISTRY PRELIMINARY EXAM
SUGGESTED SOLUTIONS (PAPER 4)

- 1 (a) (i) 1m: Fill up Table 1.1
AND both ΔT correctly calculated
AND all T & ΔT values to 1 d.p.

1m: student's $\frac{\Delta T_2}{\Delta T_1}$ within 1.8 – 2.2

(Typical result: $\Delta T_1 = 3.1\text{ }^\circ\text{C}$ and $\Delta T_2 = 6.2\text{ }^\circ\text{C}$, $\frac{\Delta T_2}{\Delta T_1} = 2.0$)

- 1 (a) (ii) 1m: Acid A is monobasic (not awarded if no explanation at all is given)

1m: Explanation based on comparison of ΔT for expt 1 vs. expt 2
AND discussion of limiting/excess reactant and no. of moles of H_2O produced

Sample answer:

		$\eta(\text{OH}^-)$ /mol	$\eta(\text{A})$ /mol	$\eta(\text{H}^+)$ /mol	$\eta(\text{H}_2\text{O formed})$ /mol
If A monobasic	expt 1	0.06 (excess)	0.015	0.015 (limiting)	0.015
	expt 2	0.03 (stoich. amt)	0.03	0.03 (stoich. amt)	0.03
If A dibasic	expt 1	0.06 (excess)	0.015	0.03 (limiting)	0.03
	expt 2	0.03 (limiting)	0.03	0.06 (excess)	0.03

A is monobasic. The amount of water formed in experiment 2 is twice that in experiment 1. This means that twice the amount of heat is liberated; as total volume is the same in both experiments, this is reflected in ΔT for experiment 2 (6.2°C) being double that in experiment 1 ($3.1\text{ }^\circ\text{C}$). If A were dibasic, the amount of water formed in both experiments is the same, and the ΔT would be the same.

- 1 (a) (iii) 1m: Calculate heat change q (may use expt 1 or expt 2)

1m: Calculate ΔH_{neut} including – sign

Sample calculation:

Heat change q = $60 \times 1.00 \times 4.18 \times 6.2 = 1555\text{ J}$

$n(\text{H}_2\text{O})$ produced = $n(\text{NaOH})$ reacted = $1.50 \times \frac{20}{1000} = 0.03\text{ mol}$

$\Delta H_{\text{neut}} = -\frac{1554.96}{0.03} = -51.8\text{ kJ mol}^{-1}$

1 (a) (iv) 1m: ΔT doubles

1m: So q doubles but will be divided by double the no. of moles of water produced, therefore no effect on ΔH_{neut}

1 (b) (i) 1m: 20.0 cm³ FA 3 + 40.0 cm³ FA 2, both volumes to 1 d.p.
AND ΔT correctly calculated
AND student's ΔT within 4.5 – 5.5

(Typical result: $\Delta T_3 = 5.0\text{ }^{\circ}\text{C}$)

1 (b) (ii) 1m: B is a weak base and compare ΔT for expt 3 (smaller) vs. expt 2

AND some energy is required to fully dissociate B (which was partially dissociated), resulting in less exothermic ΔH_{neut} , although same no. of moles of H₂O produced in expt 3 and expt 2.

2 (a) (i) 1m: Headers & units (for both time and titration recordings)
• Reject 'final volume' 'initial volume' 'amount'
• This mark is lost if final burette reading < initial or 50.00 used as initial

1m:

- transfer time in min & s to nearest whole number
- t_d correctly calculated to 1 d.p.

1m:

- burette readings, volumes to nearest 0.05 cm³ (2 d.p.)
- correct subtraction between final & initial burette readings

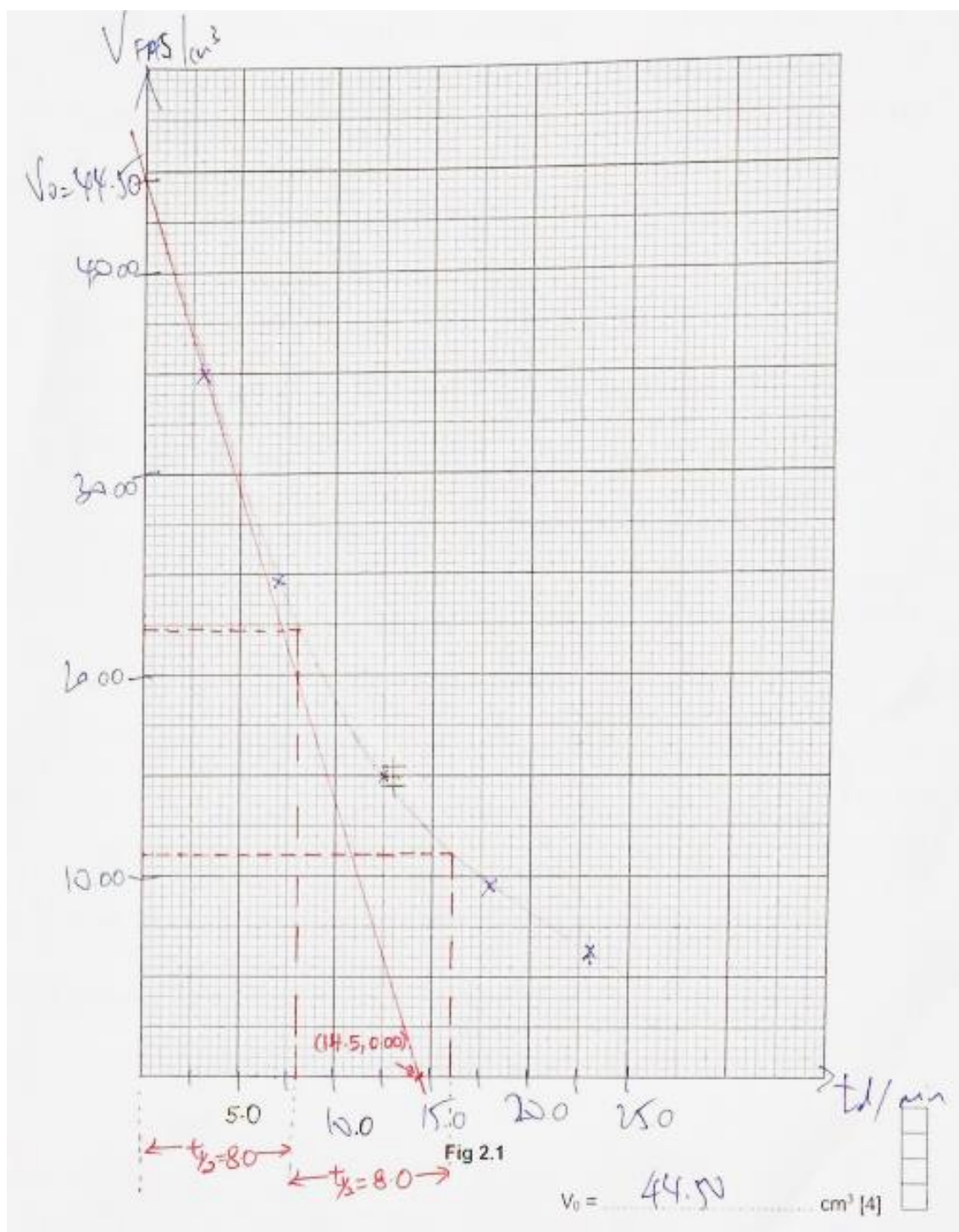
1m:

- transfer time for the first aliquot is within first 3 min of starting the reaction (accept up to 3min 56s = 3.9min)
- last aliquot not exceeding 25 min

1m: 5 sets of results

time		t_d / min	Final Burette reading / cm ³	Initial Burette reading / cm ³	Vol of FA5 / cm ³
min	s				
3	1	3.0	35.10	0.10	35.00
7	0	7.0	24.80	0.00	24.80
12	30	12.5	15.40	0.40	15.00
18	7	18.1	25.20	15.60	9.60
23	0	23.0	31.60	25.20	6.40

2 (a) (ii)



1m:

- Label x & y axes including units
- Plotted points occupy at least $\frac{1}{2}$ of graph grid in both x & y directions
- Reject awkward scale (e.g. 3:10)

1m: Plot all points ($\pm \frac{1}{2}$ small square allowance)

1m: Draw best-fit smooth curve and extend it to $t_d = 0.0$ min

1m: Read V_0 correctly ($\pm \frac{1}{2}$ small square allowance)

2 (b) (i)

$$\text{gradient} = \frac{44.50 - 0}{0 - 14.5} = -3.0690 \approx -3.07 \text{ cm}^3 \text{ min}^{-1}$$

1m: Draw tangent at $t_d = 0.0$ min, tangent line must touch the curve at $t_d = 0.0$

1m: Read coordinates correctly ($\pm \frac{1}{2}$ small square allowance)

AND Calculate gradient, include negative sign

2 (b) (ii) 1m:

$$\begin{aligned} \frac{-3.0690 \text{ cm}^3 \text{ MnO}_4^-}{1 \text{ min}} &= \frac{-\frac{3.0690}{1000} \text{ dm}^3 \times 0.0200 \text{ mol dm}^{-3}}{1 \text{ min}} \\ &= -6.1379 \times 10^{-5} \\ &\approx -6.14 \times 10^{-5} \text{ mol min}^{-1} \end{aligned}$$

If absence of negative sign has already been penalised earlier, no further penalty here.

2 (b) (iii) $2\text{MnO}_4^- + 5\text{H}_2\text{O}_2 + 6\text{H}^+ \rightarrow \text{products}$

$$\begin{aligned} \text{1m: } n(\text{H}_2\text{O}_2) \text{ decomposed per minute} &= 6.1379 \times 10^{-5} \times \frac{5}{2} \\ &= 1.5345 \times 10^{-4} \\ &\approx 1.53 \times 10^{-4} \text{ mol} \end{aligned}$$

1m:

$$\text{rate of change of } [\text{H}_2\text{O}_2] = \frac{-1.5345 \times 10^{-4}}{10/1000} = -0.0153 \text{ mol dm}^{-3} \text{ min}^{-1}$$

If absence of negative sign has already been penalised earlier, no further penalty here.

1m: sf for 1(a)(iii), 2(b) and units for 1(a)(iii)

2 (b) (iv)	Volume FA 3 /cm ³	half-life /min
	44.50 → 22.25	8.0 – 0.0 = 8.0
	22.25 → 11.125	16.0 – 8.0 = 8.0

Agree. The half-lives are constant at 8.0 minutes. Since the $t_{1/2}$ are constant, it is a 1st order reaction with respect to H_2O_2 .

1m: Quote at least two half-life values or label on the graph

1m: Conclusion based on whether half-life is (approximately) constant or not

2 (c) (i) A brown solution was initially observed on adding orange $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$.

Then vigorous effervescence of a colourless gas which relights a glowing splint. The gas is O₂

An orange/yellow solution was observed at the end.

$\frac{1}{2}$ m for each underlined observation $\Rightarrow$ 2m

2 (c) (ii) 1m: Suggests $[\text{Fe}(\text{H}_2\text{O})_5(\text{O}_2\text{H})]^{2+}$ as the brown intermediate (accept $[\text{Fe}(\text{H}_2\text{O})_5\text{O}]^{3+}$) 1m: suggests orange/yellow $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ regenerated

2 (c) (iii) 1m: A brown intermediate was formed, showing that $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ participated in the catalysed pathway, as shown in step 1

1m: Vigorous effervescence of oxygen indicated an increase in the rate of reaction

1m: The catalyst was regenerated as seen by the return of the yellow/orange colour at the end and as shown in step 3

- 3 (a)
1. White solid dissolves to give a colourless solution.
 2. Effervescence/bubbles of gas
 3. Gas gives white ppt when bubbled through limewater. Gas is CO_2 .
 4. Black residue
 5. Colourless/pale pink/yellow filtrate
 6. Filtrate gives off-white/light brown ppt with $\text{NaOH}(\text{aq})$
 7. Ppt is insoluble in excess
 8. Ppt rapidly turns brown on contact with air

1m for every two observations $\Rightarrow$ 4m

3 (b) 1m: No more effervescence is observed when FA 8 is added.

3 (c) 1m: Oxidising agent, as ethanedioate/ethanedioic acid was oxidised to carbon dioxide.

3 (d) (i) 1m: Mn^{2+} , as off-white ppt rapidly turns brown on contact with air, insoluble in excess

3 (d) (ii) 1m: MnO_2

1m: FA 8 acted as an oxidising agent, so itself must have been reduced from a higher oxidation state of Mn to Mn^{2+} , and MnO_2 , the oxide with Mn in +4 oxidation state, has $M_r = 86.9$, which is less than 140.

4 (a) There are 10 marking points (M1 – M10)

No. of correct marking points	0–1	2–3	4–5	6–7	8–9	10
No. of marks scored	0	1	2	3	4	5

M1: Separate measuring cylinders (or burettes) to measure volume of each liquid (aqueous ethanoic acid, cyclohexane, deionised water).

M2: Shake the mixture in the separating funnel (or invert the separating funnel a few times to mix the liquids). Then leave the mixture to stand / to equilibrate / to separate into two layers.

M3: Mixture 1 contains 50 cm^3 aq. ethanoic acid and 50 cm^3 cyclohexane.

AND at least a total of four mixtures stating the volumes of aq. ethanoic acid, cyclohexane and deionised water.

(The top of pg 13 says 'Using the information given'. The bottom of pg 12 says 'using different volumes of aq. ethanoic acid, and adding water such that the total volume of the aq. and cyclohexane layers is kept constant'. So you should stick to 50 cm³ cyclohexane for all mixtures.)

M4: Total volume of aq. layer and cyclohexane layer kept constant.

(M4 is lost if student mixed aq. ethanoic acid and water, never add cyclohexane / mixed water and cyclohexane, never add aq. ethanoic acid.)

M5: Turn the tap of the separating funnel to drain out most of the lower aqueous layer into a beaker.

(M5 is lost if student uses the wrong apparatus e.g. volumetric flask / conical flask to prepare the mixture. Please use separating funnel.)

(Some students poured aq. ethanoic acid/cyclohexane into a beaker/conical flask, *then* transferred the mixture into the separating funnel (instead of using the separating funnel directly). If you do this, some liquids will stay in the beaker/conical flask $\Rightarrow$ loss of reagents $\Rightarrow$ M5 lost.)

M6: Pipette out 10.0 cm³ of the aqueous layer from the beaker (after M5).

(Reject use of dropping pipette here. The dropping pipette (it's actually just a plastic dropper) was used in Question 2 because the bubbling H₂O₂ makes it difficult to use a 10.0 cm³ or 25.0 cm³ glass pipette.)

M7: Place the 10.0 cm³ aqueous aliquot into a conical flask.

M8: Add 2 or 3 drops of thymol blue to the aq. aliquot.

M9: Titrate the aq. aliquot with aq. NaOH from a burette.

AND record initial burette reading and final burette reading.

M10: Titrate until yellow turns green.

(The cyclohexane layer cannot be titrated with NaOH(aq). NaOH is an ionic compound, its OH⁻ ions have no favorable interactions with the non-polar cyclohexane molecules. So the OH⁻ ions will not 'enter' the organic solvent to react with the CH₃CO₂H molecules.)

SAMPLE WRITE-UP:

The equilibrium mixtures to be prepared are stated below.

mixture	Volume /cm ³			Titration of 10.0 cm ³ aq. aliquot against NaOH(aq)		
	aq. ethanoic acid	deionised water	cyclohexane	Final burette reading /cm ³	Initial burette reading /cm ³	Volume of NaOH(aq) needed /cm ³
1	50	0	50			
2	40	10	50			
3	30	20	50			
4	20	30	50			
5	10	40	50			

1. Use separate measuring cylinders to measure 50 cm³ of aq. ethanoic acid and 50 cm³ of cyclohexane. Add both liquids into the separating funnel.
2. Stopper and shake the separating funnel to mix the liquids. Leave the separating funnel to stand for a few hours.
3. Turn the tap of the separating funnel to drain out most of the lower aqueous layer into a beaker.
4. Pipette 10.0 cm³ of the aqueous layer, from the beaker in step 3, into a conical flask. Add 2 or 3 drops of thymol blue indicator.
5. Titrate the aqueous aliquot with (or against) the NaOH(aq) from a burette until the solution just changes from yellow to green. Record the initial and final burette readings.
6. Repeat steps 1 to 2 to prepare the other four mixtures. Use a third measuring cylinder to measure the volume of deionised water. Then repeat steps 3 to 5 for the titration of the aqueous aliquot.

4 (b)

1m: n(NaOH) needed for titration

1m: [CH₃CO₂H] in aqueous layer

1m: subtraction working to obtain n(CH₃CO₂H) in cyclohexane layer including scale-up-working for n(CH₃CO₂H) in the aqueous layer

1m: [CH₃CO₂H] in cyclohexane layer

(equilibrium mixture 1: 50 cm³ aq. CH₃CO₂H + 50 cm³ cyclohexane)

(V cm³ NaOH(aq) needed for titration of 10.0 cm³ aq. layer)

$$n(\text{NaOH}) = \frac{V}{1000} \times 0.100 = a \text{ mol}$$

$$n(\text{CH}_3\text{CO}_2\text{H}) \text{ in } 10.0 \text{ cm}^3 \text{ aq. layer} = a \text{ mol}$$

$$[\text{CH}_3\text{CO}_2\text{H}] \text{ in aq. layer} = \frac{a}{10/1000} \text{ mol dm}^{-3}$$

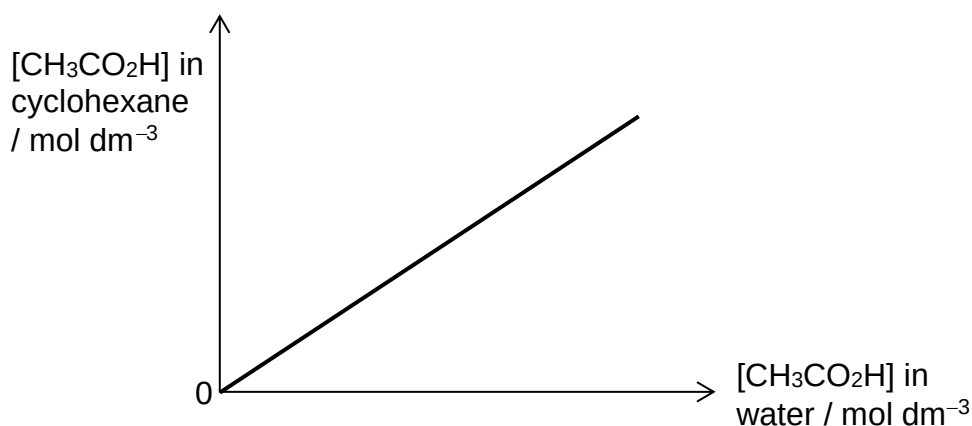
$$n(\text{CH}_3\text{CO}_2\text{H}) \text{ in } 50 \text{ cm}^3 \text{ aq. layer} = a \times \frac{50}{10} = b \text{ mol}$$

$$n(\text{CH}_3\text{CO}_2\text{H}) \text{ in the original } 50 \text{ cm}^3 \text{ aq. acid} = \frac{50}{1000} \times 0.250 = 0.0125 \text{ mol}$$

$$n(\text{CH}_3\text{CO}_2\text{H}) \text{ in cyclohexane layer} = 0.0125 - b \text{ mol}$$

$$[\text{CH}_3\text{CO}_2\text{H}] \text{ in cyclohexane layer} = \frac{0.0125 - b}{50/1000} \text{ mol dm}^{-3}$$

- 4 (c) (i) **1m:** draw a straight line, positive gradient, joins origin (steepness doesn't matter)



- 4 (c) (ii) **1m:** K = gradient of this line

- 4 (d) **1m:**
 Ethanoic acid is a weak acid that ionises partially / dissociates partially in water to give CH_3CO_2^- and H^+ ions.
 (Reject 'hydrolyses' / 'hydrolysis')
 or Ethanoic acid molecules may form dimers in the cyclohexane solvent.
 Therefore, the assumption is not entirely true.